

Satellite Image Dehazing Via Masked Image Modeling and Jigsaw Transformation

Guisik Kim¹, Choongsang Cho¹, and Junseok Kwon²

¹ Korea Electronics Technology Institute, Korea

² School of Computer Science and Engineering, Chung-Ang University, Seoul, Korea
specialre@naver.com, ideafisher@keti.re.kr, jskwon@cau.ac.kr

Abstract. In this paper, we successfully apply masked image modeling (MIM) to the dehazing process for satellite images, introducing a novel dehazing method. Initially, we investigate why MIM does not effectively function as a self-supervised learning method for low-level vision tasks and fails to yield improved performance. Subsequently, we propose two solutions to address this issue. Furthermore, we introduce an augmentation technique that enhances both locality and non-locality in puzzle images through jigsaw transformations, resulting in improved accuracy. Experimental results show that our method outperforms other state-of-the-art methods, including approaches based on visual transformers.

Keywords: Satellite image dehazing · Masked image modeling · Jigsaw transformation

1 Introduction

Recently, satellite images are utilized across numerous domains, including geoscience, agriculture, forestry, urban planning, disaster monitoring, national security, import/export industries, meteorology, and everyday life. Computer vision techniques have been applied to enhance these images, enabling various satellite image applications. However, the effectiveness of existing computer vision methods can be hindered by haze or cloud cover, which frequently obscures satellite imagery. According to data from Terra and Aqua satellites, approximately 67 percent of the Earth’s surface is obscured by clouds [21]. While atmospheric correction processes can partially mitigate the effects of the thin atmosphere, clouds, haze, and other atmospheric phenomena still persist in satellite imagery. Consequently, satellite images must undergo several correction processes to generate images suitable for conventional computer vision methods to function effectively.

To implement advanced correction processes, our proposed method leverages pre-trained models based on self-supervised learning. In particular, it utilizes masked image modeling (MIM) to address various challenges encountered when applying this technique to low-level vision tasks.

1.1 Motivation

MIM demonstrates high generalization performance as a self-supervised learning method [16, 39], aligning with recent trends in training large-scale language

models. While originally derived from language modeling, it has recently begun to be applied to the image domain in high-level vision tasks, bringing about considerable performance improvements. In high-level vision tasks, networks learn the overall representation of an image by restoring a randomly masked image to its original form. Consequently, MIM proves valuable for enhancing performance in downstream tasks by comprehensively understanding and learning image representations, surpassing pre-training models for classification [16, 39].

In contrast, in low-level vision tasks like satellite dehazing, MIM does not effectively function as a self-supervised learning method. The fundamental difference between low-level vision and high-level vision lies in whether the focus is on individual pixels or on the overall object. MIM for high-level vision tasks is tailored to learn the representation of entire objects, making it less suitable for low-level vision tasks where pixel-level detail is crucial. Therefore, in this paper, we raise the question: *"How can we effectively apply MIM to low-level vision tasks such as satellite dehazing?"*

1.2 Problem Statements

This paper addresses the aforementioned question by considering two points.

1. **Convolutions:** We need to utilize convolutions that capture information from adjacent pixels. In image restoration tasks, where the goal is to restore each pixel, focusing on small details is crucial and often more significant than considering the overall context of the object. The visual transformer (ViT) typically captures long-range dependency characteristics. Thus, despite the suitability of the visual transformer structure for implementing masked image modeling, ViT does not always lead to performance improvement.
2. **Small mask patches:** We need to use a smaller mask patch size. In high level tasks such as classification, object detection, and segmentation, where the main target of the photo is clear and the overall context is considered important, relatively larger masking was possible (*e.g.* a 32×32 patch size in SimMIM [39]). However, in low-level vision tasks, large masks result in a significant loss of detail because they cannot effectively restore the surrounding context. In contrast, employing a small mask size and convolutions that consider adjacent pixels enables the network to learn the local context of the image. With these two considerations, we can effectively utilize MIM for satellite dehazing.

1.3 Contributions

The main contributions of the proposed method can be summarized as follows.

- We present a novel dehazing method for satellite images by successfully applying masked image modeling to the dehazing process.
- We introduce Jigsaw puzzle data augmentation for dehazing. Jigsaw puzzle augmentation can reduce the spatial connectivity of an image without altering the pixels, thereby facilitating the learning of local characteristics. This

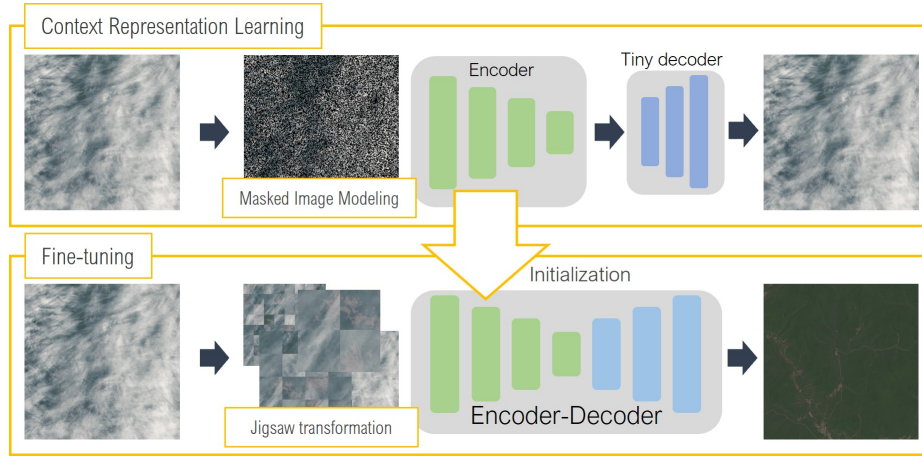


Fig. 1: Overview of the proposed method. The proposed method can be mainly divided into two parts. It comprises a pretraining stage that conducts context representation learning via haze reconstruction using MIM, followed by a fine-tuning stage where a network initialized with context representation parameters is trained for dehazing. During the fine-tuning stage, the proposed jigsaw transformation helps in obtaining better dehazing results.

approach is particularly suitable for low-level vision tasks that emphasize detail restoration. Furthermore, it behaves similarly to non-local learning near the boundaries of puzzle pieces. This enhancement is achieved not by improving convolution operations, but rather through physical transformations of the data, eliminating the need for complex computations.

2 Related Work

2.1 Image Dehazing

Dehazing has a long history in low-level vision. The research trajectory has evolved from prior-based methods [4, 14, 17] to deep learning-based methods [5, 22, 27, 31, 33]. Recent studies in haze removal primarily concentrate on enhancing performance through network design, with many utilizing transformers [15, 32, 34]. Due to the lack of paired datasets containing images with and without haze, researchers have frequently relied on datasets generated with synthetic haze [23] or fog generators [1–3]. FFANet [31] can be considered a fundamental method based on CNNs with a feature attention block. DMF²-Net [9] incorporated a pre-training model not commonly used in haze removal and combines multiple outputs. [18, 20] proposed distillation from a pre-training model. AECR-Net [38] effectively integrated contrastive learning, while DeHamer [15] applied transformers to dehazing. DehazeFormer [34] also demonstrated high dehazing performance as a transformer-based method. Finally, MB-TaylorFormer [32] achieved the highest performance by leveraging Taylor expanded self-attention.

With the rise in the number of satellite launches, research on haze removal in satellite images has become increasingly active. In satellite imagery, apart

from clouds with specific shapes, haze-like characteristics can be observed when they are widespread. Consequently, cloud removal research and dehazing field research have often overlapped in satellite imagery [24, 29, 34]. In [34], a satellite image haze dataset was introduced, leveraging the cirrus band to construct the dataset. This dataset synthesized cirrus clouds actually captured by satellites, resulting in very natural shapes and distributions. Thus, RSHaze [34] currently stands as the most advanced dataset in this domain. EMPF-Net [36] proposed an encoder-free network, while DFPN [19] achieved high performance using a simple method of dividing haze images into low- and high-frequency components.

In contrast, the proposed method focuses on a learning methodology through MIM and augmentation using jigsaw transformation, rather than network improvement. These two methods enable the network to enhance its performance by considering the understanding of haze and learning locality and non-locality.

2.2 Masked Image Modeling

MIM [16, 39] stems from the self-supervised learning technique of masked language modeling [10, 43] and has been applied to the field of computer vision. Through experiments, it has been proven that pre-training models restoring masked regions contribute to performance improvements in other high-level vision tasks [16, 39]. MIM can be tailored to fit the vision transformer [12] architecture, designing an encoder to restore masked regions [39] or enhancing the network’s inference capabilities by training on the unmasked areas [16]. While most designs are transformer-based, there are also studies incorporating convolution [35, 37]. The issue of significant distribution gaps between masked and original images when using convolution, leading to substantial zeroing-out, was addressed by implementing sparse convolution [26]. Masking was applied to the final encoder layer in [37]. Masked image modeling was directly applied to image processing without any specific modifications [16].

The proposed method is designed to focus more on details using small patches and convolution. This adjustment makes MIM more suitable for dehazing tasks.

2.3 Data Augmentation for Low-level vision

In high-level vision, data augmentation techniques such as jitters, transformation, cutMix [41], cutOut [11], and random erase [42] are considerably more diverse than those used in low-level vision tasks. Typically, image restoration has only used flip series such as horizontal flip and vertical flip. CutBlur [40] is an augmentation technique specifically tailored to suit the characteristics of super-resolution. However, this approach is difficult to apply to various algorithms because it is designed for specific tasks.

We propose a jigsaw transformation technique that can be applied to various image restoration fields without damaging the image. While there are existing studies that have utilized jigsaw-related approaches [7, 30], the jigsaw transform is a novel approach that has not been previously used in data augmentation for low-level vision, yet it can effectively capture both locality and non-locality.

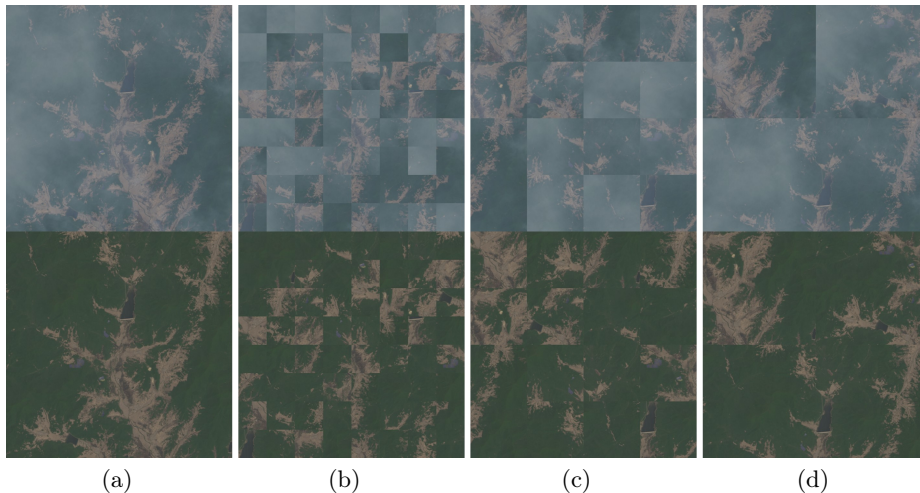


Fig. 2: Jigsaw transformation example. (a) Haze image, (b), (c), and (d) are jigsaw transformation of 64 pieces, 16 pieces, and 4 pieces, respectively.

3 Proposed Method

In this section, we delve into the proposed method in detail. First, in Section 3.1, we revisit the design of MIM for context representation learning. Then, in Section 3.2, we introduce the pretraining method using MIM. In Section 3.3, we discuss the fine-tuning process, aiming for our ultimate goal of dehazing. Finally, in Section 3.4, we explain the proposed jigsaw transformation, examine its effects, and discuss its specifics.

3.1 Revisiting MIM For Context Representation Learning

We examine the reasons why applying a typical masked autoencoder (MAE) to low-level vision algorithms is challenging. MAE masks random patches and reconstructs the missing patches, which can lead to the loss of details such as edges, textures, and structures. Consequently, the application of a typical MAE may have a detrimental effect on image restoration performance. In addition, while many recent methods are designed with a ViT structure, transformers do not capture local details as effectively as convolutions, which are known for their ability to handle long-range dependencies. For example, the model [13] was not designed to prioritize details. DegAE [28] demonstrated that applying MAE to ViT for low-level vision significantly reduces PSNR performance compared to state-of-the-art methods, leading to over-smoothing and artifacts.

Based on these findings, the proposed method applies a convolution operation to MIM. However, when the mask size exceeds the filter size of the convolution layer, traditional convolutional methods do not function optimally. Therefore, recent works [35, 37] have explored the approach of applying sparse convolutions. In contrast to these methods, our method reduces the mask size and encourages the model to learn context representations capable of restoring the masked areas by leveraging information from adjacent pixels.

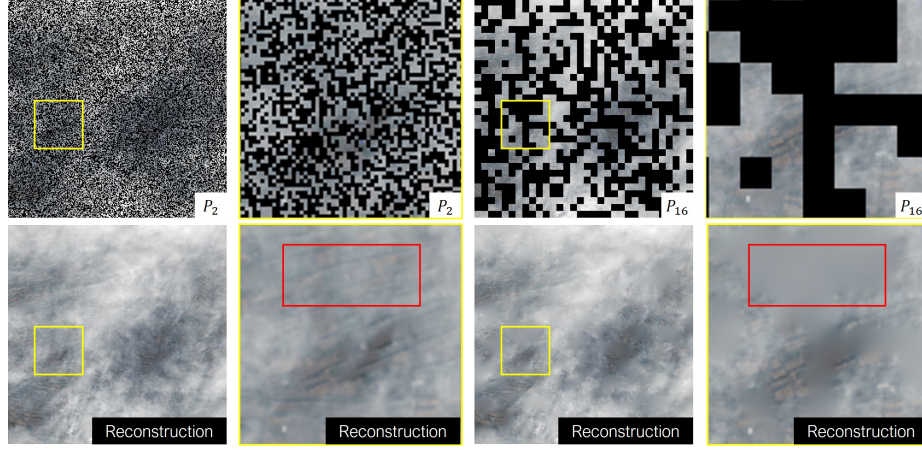


Fig. 3: Comparison according to the size of the mask patch. The first and third columns show the masking images and haze reconstruction results for mask size $2(P_2)$ and mask size $16(P_{16})$, respectively. Examining the enlarged red box area in the second and fourth columns, the image on the left, with a small mask patch size, is successfully restored by referencing the surroundings. However, with large mask patches, convolution cannot adequately reference the surroundings, inducing over-smoothing.

3.2 Pretraining MIM stage

We use NAFNet [6] as our baseline network. NAFNet has a simpler convolutional structure and lower computational complexity compared to the self-attention mechanism in basic transformers. Please note that the proposed method does not aim to introduce new network structures; instead, we simply modify NAFNet into a symmetric structure and use it as a baseline. For detailed network description and figures, please refer to [6].

As NAFNet is symmetrically modified, each level of the encoder consists of [1, 2, 4] blocks, and the middle block, which serves as the symmetry point between the encoder and decoder, also contains 4 blocks. Downsampling is performed using convolutions with kernel size 2 and stride 2. Then, representation learning for both haze and the context behind the haze can be formulated as follows.

$$\hat{I} = D_t(E(I_M)), \quad (1)$$

where I_M represents randomly masked haze images. After passing through encoder E and tiny decoder D_t , we can obtain reconstructed haze images $\hat{I}$. As previously described, the degree of restoration is determined by the size of the masking, as illustrated in Fig. 3. The tiny decoder follows a simple architecture where conv-relu-bilinear upsampling is symmetrically arranged according to the downsampling factor. Then, we compare the restored haze image $\hat{I}$ with the masked haze image I_M only for the masked regions, similar to SimMIM [39].

$$\mathcal{L}_m = \frac{1}{\Omega(I_M)} \|\hat{I}_M - I_M\|_1, \quad (2)$$

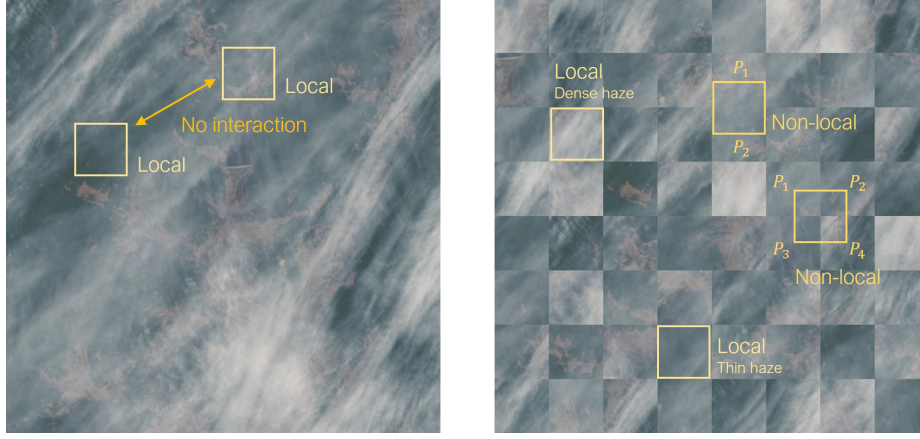


Fig. 4: Illustration of the operating principle of the Jigsaw transformation

where $\hat{I}_M$ denotes the restored haze image that contains only the masked regions, and $\Omega(\cdot)$ is the number of masked pixels.

3.3 Fine-tuning Stage

During the pretraining MIM stage, the trained encoder leverages adjacent pixels to acquire a context representation for restoring the masked areas. This process is performed in the haze domain, aiming to enhance the network’s understanding of both haze and the context behind it. To effectively utilize this pretrained encoder for haze removal, fine-tuning is necessary. Fine-tuning utilizes the overall architecture of NAFNet, as illustrated in Fig.1, where a decoder symmetric to the encoder rather than a tiny decoder is employed to enhance the haze removal capability. In fine-tuning, as the objective is haze removal, we compare the haze-free image J with the final result passed through the encoder-decoder $D(E(I))$.

$$\mathcal{L}_1 = \|D(E(I)) - J\|_1. \quad (3)$$

3.4 Learning Locality and Non-Locality via Jigsaw Transformation

Unlike transformers, convolutional networks are fundamentally specialized in capturing local information. As the layers of the network deepen, the receptive field expands to encompass global characteristics. Images inherently possess spatial continuity and are represented similarly to the information received by the brain through human eyes. Therefore, our brain can perceive objects and backgrounds based on this visual information. As shown in [30], when a network learns to solve a jigsaw puzzle, it tends to improve its representation learning of objects. This is analogous to how young children and the elderly engage in cognitive training through puzzles. In this context, learning by breaking spatial connectivity can also aid in dehazing, with a focus on restoring local pixels.

Fig.4 shows the jigsaw transformation. The typical image on the left does not undergo the jigsaw transformation. If a convolution is performed on this, no interaction can be expected between the box areas. In contrast, in the jigsaw

Table 1: (Ablation Study) Comparison of performance based on component-wise evaluation and masking ratio. The best results are written in boldface.

	Baseline	Jigsaw	MIM	$\gamma=0.9$	$\gamma=0.75$	$\gamma=0.5$
PSNR	40.01	40.28	40.06	40.36	40.31	40.43
SSIM	0.963	0.973	0.973	0.963	0.973	0.974

transformation image on the right, the network can learn about the local characteristics that maintain the continuity of the image within each puzzle piece. However, as the layer deepens and the receptive field enlarges, the area corresponding to the boundary of the puzzle piece can be part of a completely different region lacking spatial continuity. Then, the network exhibits a tendency towards augmentation, enabling it to consider both local and non-local information.

The advantages of augmentation through jigsaw transformation can be summarized as follows. 1) The pieces within the jigsaw transformation contain local information that can effectively leverage the benefits of convolution. 2) Near the boundary of the piece of the jigsaw transformation and in deep layers where the receptive field is enlarged, information from non-adjacent areas, similar to a non-local network, can be utilized for learning. 3) As the local and non-local augmentation occur simultaneously, the network can learn about diversity and effectively fulfill its role from the perspective of augmentation. Then, the Jigsaw transformation $Jigsaw(\cdot)$ can be expressed as follows.

$$I_j = Jigsaw(I, N), \quad (4)$$

where I_j denotes the j -th piece and N is the number of pieces into which the image is divided. In Fig. 2(b), the image is divided into N^2 64 pieces with $N = 8$. Although the self-attention of the transformer can consider the relationship between all areas, the proposed method can capture local information more effectively than the transformer. This difference stems from the earlier explanation regarding the rationale for utilizing convolution in conjunction with MIM.

4 Experiments

We utilized the RSHaze [34], RICE [25], and RSID [8] datasets for satellite images, including aerial images. RSHaze utilizes Landsat-8 satellite images, which has been introduced by DehazeFormer [34]. It is the largest and most sophisticated dehazing dataset from existing satellite images, utilizing a satellite cirrus band. The dataset comprises 51,300 images for training and 2,700 images for testing. The RICE dataset [25] consists of 500 paired images collected from Google Earth. On the other hand, the RSID dataset [8] includes 1,000 paired images of haze and clean images. While other datasets are predominantly distributed in mountains and cities, RSID is characterized by containing various objects, mainly ships and airplanes.

4.1 Implementation Details

The proposed method was implemented in PyTorch and used an NVIDIA A6000 GPU. We employed the AdamW optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and the

Table 2: (Ablation Study) Comparison of haze reconstruction and fine-tuning performance. The best results are written in boldface.

	$P_{16}, \gamma=0.5$		$P_8, \gamma=0.5$		$P_2, \gamma=0.5$		$P_2, \gamma=0.75$		$P_2, \gamma=0.9$	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Haze Reconstruction	35.78	0.921	37.73	0.943	41.44	0.975	38.46	0.948	35.73	0.902
Fine-tuning	40.28	0.973	40.26	0.973	40.43	0.974	40.31	0.973	40.36	0.973

learning rate was reduced from $2e-4$ using a cosine annealing strategy. Random crop patches of size 192×192 were utilized for both context representation learning and subsequent fine-tuning. For the RSID dataset, which is relatively small in size, we utilized patches of 128×128 . Considering the dataset sizes, we employed a batch size of 16 for the RSHaze dataset and 1 for the RICE and RSID datasets. We empirically set 900k iterations as the final learning duration.

For basic augmentation, excluding the Jigsaw transformation, we applied horizontal and vertical flips. The Jigsaw transformation operated with a probability of 0.5 and was randomly divided into N^2 pieces, where $N = [2, 4, 8, 16]$. Afterward, random cropping was applied to ensure the presence of various boundaries and patch numbers. In the pretraining MIM stage, the point at which the peak signal-to-noise ratio (PSNR) score of the haze reconstruction on the validation set reached its peak was saved and used as the initial weight for fine-tuning. We used the validation set from a subset of the test set. Because the RSHaze dataset has 9 types of haze per image, it consists of 9 images. The highest validation score for the proposed haze reconstruction was 43.14dB. The masking ratio γ was experimentally set to 0.5.

4.2 Ablation Study

To demonstrate the effectiveness of our proposed method, we conducted several experiments. Table 1 shows the results of component-wise experiments and experiments on the masking ratio γ of our proposed method. Firstly, for the component-wise case, we performed MIM and Jigsaw transformations separately. Both the proposed Jigsaw and MIM recorded higher PSNR values than the baseline network. When $\gamma=0.5$, it represents the experimental results of our proposed method applying both Jigsaw transformation and MIM, denoted for comparison with varying γ values. We can confirm that the jigsaw transformation and MIM each contributed to higher performance through mutual complementary learning. As evidenced by the structural similarity index (SSIM) score, all results showed a very high value of over 0.973, indicating that the performance of the dataset itself is somewhat saturated. Nevertheless, the fact that the PSNR value has increased by 0.42 dB compared to the baseline suggests a significant improvement cause by the proposed method.

Table 2 presents the experiments according to different mask patch sizes of MIM (p) and the masking ratios (γ). We conducted experiments by increasing the mask patch based on the proposed $P=2$ and $\gamma=0.5$, and by increasing the masking ratio. As the size of the patch increases, as explained earlier, the convolution cannot perform reconstruction by referring to adjacent pixels, resulting

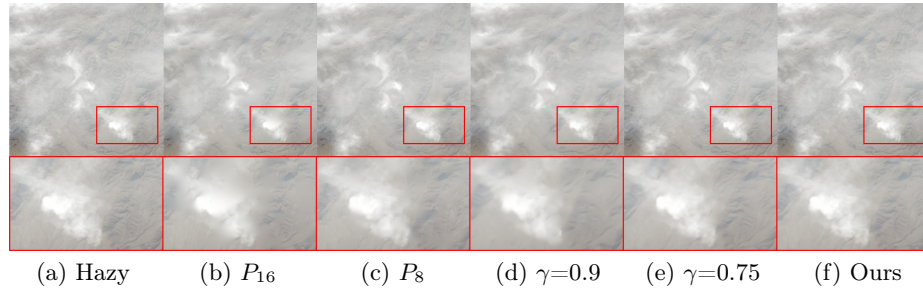


Fig. 5: (Ablation Study) Haze reconstruction results according to different MIM configurations.

in decreased PSNR and SSIM values. Similarly, when the patch size is fixed and the masking ratio increases, the number of adjacent pixels that can be referred to decreases, leading to decreases in both PSNR and SSIM values. This is observed in the haze reconstruction experiment. Fig. 5 presents the qualitatively results of the aforementioned experiment. Consistent with the overall trend observed in the quantitative evaluation of haze reconstruction, we can confirm that when the patch size is large or the masking ratio is high, the result tends to be over-smoothed, leading to inadequate restoration of details. In the fine-tuning stage, a comparative analysis was conducted on the final dehazing outcomes by initializing the weight of the learned encoder, as shown in Table 2. Because the encoder also undergoes tuning during this process, while the final PSNR and SSIM results may not appear markedly different compared to the decrease in haze reconstruction, it is notable that the proposed small masking size and $\gamma=0.5$ exhibit the highest results.

Fig. 6 illustrates the features passed through the first encoder block of both the proposed method and the baseline algorithm. The visualization represents the average result of all channels. Typically, in network learning, low-level features are represented at lower levels, while high-level features emerge as the network progresses to higher levels. In the baseline results shown in Fig. 6(b), we can observe that the information about the haze, a low-level feature, is clearly visible on the feature map. In contrast, the proposed method prominently reveals information about the image context rather than emphasizing the features of the haze. This tendency is consistent across the second rows, suggesting a characteristic of the proposed method.

4.3 Comparison with Other Methods

For the quantitative and qualitative comparison, we conducted experiments with various algorithms on the RSHaze [34], RICE [25], RSID [8] datasets. Since algorithms solely for satellite image dehazing have not been extensively researched yet, we made some comparisons with ground-level dehazing algorithms such as DCP [17], DehazeNet [5], AOD-Net [22], GDN [27], FFA-Net [31], which are representative methods in the field. Additionally, DehazeFormer [34] demonstrates

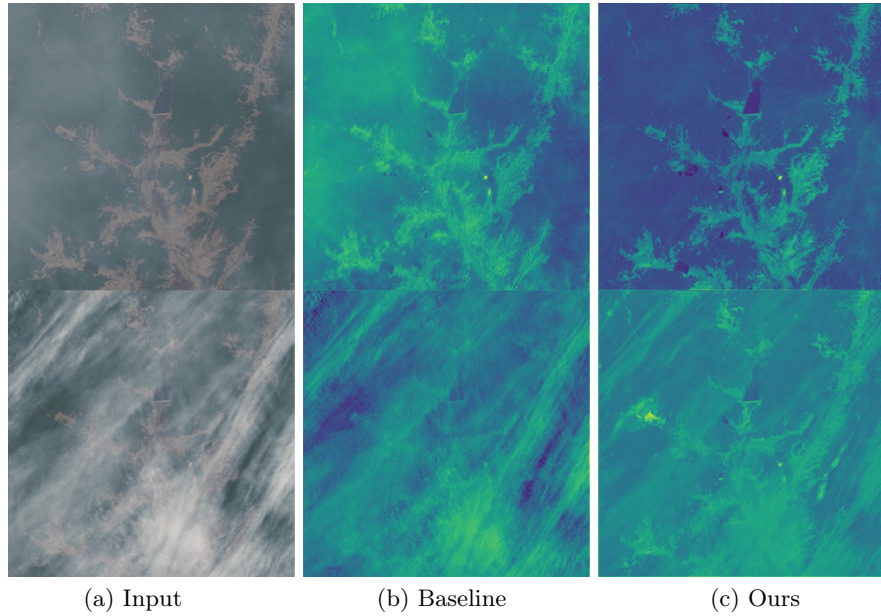


Fig. 6: (Ablation Study) Validation of the context representation learning effect through feature analysis.

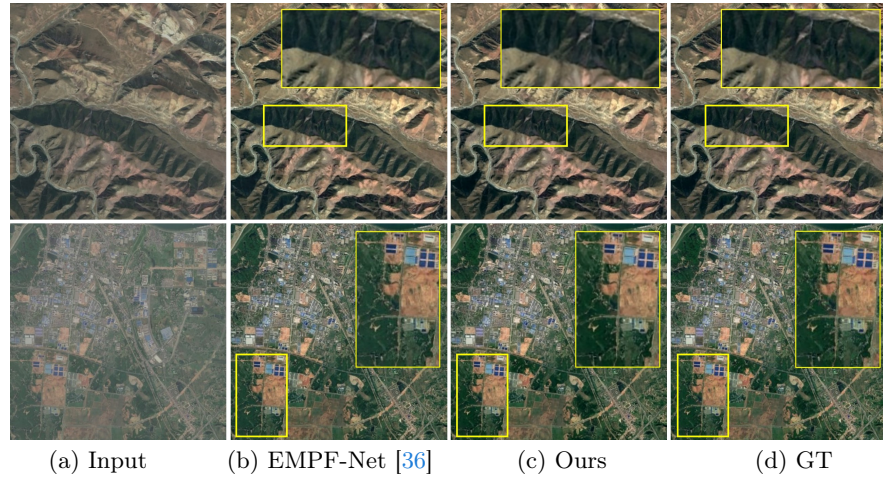
high performance not only in ground-level dehazing but also in satellite image dehazing. Moreover, EMPF-Net [36] is one of the most recent algorithms and has exhibited exceptional performance in satellite image dehazing.

Table 3 presents the comprehensive experimental results of the proposed method and other algorithms. To ensure a fair evaluation, the result images and the numbers in the table were obtained from the official code and the original papers. In the RICE dataset, both EMPF-Net and the proposed method exhibit remarkably high results compared to other algorithms, with the proposed method achieving the highest performance in terms of PSNR and EMPF-Net achieving the highest performance in terms of SSIM. Since SSIM has already recorded a score of over 0.98, indicating saturation in performance, a difference of 0.003 can be considered very minor. However, despite the saturated SSIM scores, the fact that the proposed method achieves approximately 1dB higher in PSNR can be viewed as a significant difference. Fig. 7 contain qualitative results on the RICE dataset. The RICE dataset is characterized by global haze, and both EMPF-Net and our method demonstrate decent performance in dehazing. However, a slight difference in tone can be observed. The proposed method exhibits slightly higher saturation than EMPF-Net and appears more similar to the ground truth.

The RSHaze dataset [34] is the most extensive and meticulously constructed dataset, making it highly reliable. DehazeFormer [34], EMPF-Net [36], and the proposed method all achieved high SSIM scores of over 0.97. In terms of PSNR, only EMPF-Net [36] and the proposed method achieved scores exceeding 40. Similar to the RICE dataset, a difference of 0.003 in SSIM can be considered

Table 3: Quantitative comparison using the RS-Haze [34] and RICE datasets [25] in terms of PSNR and SSIM. The best results are boldfaced.

Methods	RICE		RS-Haze		Overhead	
	PSNR	SSIM	PSNR	SSIM	#Param	MACs
DCP [17]	16.03	0.697	17.44	0.822	-	-
DehazeNet [5]	23.74	0.865	23.16	0.816	0.009M	0.525G
AOD-Net [22]	19.84	0.740	24.90	0.830	0.002M	0.115G
GDN [27]	30.01	0.966	36.40	0.960	0.956M	21.49G
FFA-Net [31]	31.78	0.954	39.39	0.969	4.456M	287.8G
DehazeFormer [34]	31.88	0.962	39.57	0.970	1.283M	13.13G
EMPF-Net [36]	35.96	0.987	40.05	0.977	0.427M	1.589G
Ours	36.92	0.984	40.43	0.974	13.06M	16.48G

**Fig. 7: Qualitative comparison using the RICE dataset [25].**

very minor, but a difference of about 0.4dB at a very high level where the PSNR score exceeds 40 is considered significant. Fig. 8 shows the qualitative results for the RS-Haze dataset. EMPF-Net exhibited results that was not as good compared to its score, while DehazeFormer and the proposed method demonstrated high haze removal performance. When examining the enlarged red box area in the first row, the result of DehazeFormer shows that the color of the river was harmonized with the surroundings and transformed into a dark green hue. In contrast, the proposed method did not exhibit this phenomenon and it maintained the color of the river intact. In the second and third rows, it is demonstrated how much the line appearing in the mountain range is preserved even after dehazing. The proposed method maintained the closest resemblance to the original, while the results of other algorithms appeared as if the line is cut off. More results can be found in the **supplementary materials**.

Table 4 compares the proposed method with FFANet, which exhibited the highest performance based on convolution in other datasets, and EMPF-Net, the latest algorithm. The experimental results indicate that the proposed method demonstrated high performance in terms of both PSNR and SSIM on the RSID [8]

Table 4: Quantitative comparison using the RSID dataset [8] in terms of PSNR and SSIM. The best results are written in boldface.

	FFA-Net [31]	EMPF-Net [36]	Ours
PSNR	25.08	23.02	25.57
SSIM	0.940	0.926	0.945

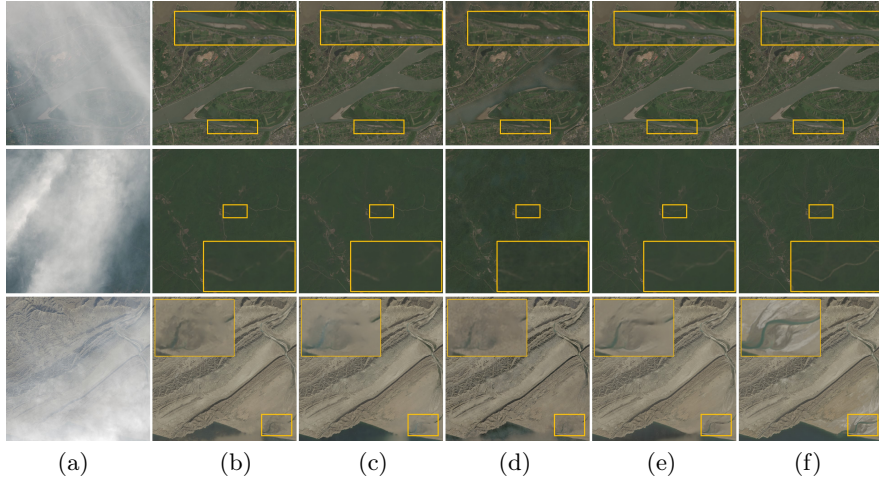


Fig. 8: Qualitative comparison using the RSHaze dataset [34]. (a) Input, (b) FFA-Net [31], (c) DehazeFormer [34], (d) EMPF-Net [36], (e) Ours, (f) GT

dataset. Fig.9 shows the qualitative results for the RSID dataset. It is evident that our method closely resembles the ground truth, with no observable artifacts.

From the overhead perspective, the proposed method has a large number of parameters, and the multiply-accumulate operations (MACs) are also recorded as higher than Dehazeformer and the encoder-free network, EMPF-Net. This is the result of applying the same specifications as the baseline network, NAFNet, and the proposed MIM and jigsaw transformation did not have any impact. Therefore, applying the proposed method as an add-on to other algorithms does not affect inference. MACs were tested at a size of 256×256 for all algorithms. Regardless of the overhead, the processing time for a 512×512 image was measured as 0.049 seconds for DehazeFormer, 0.028 seconds for the proposed method, and 0.022 seconds for EMPF-Net. These results demonstrate a negligible difference between each method.

4.4 Discussion and Limitation

Fig.10 illustrates a failure case that occurs not only in the proposed method but also in most algorithms. Very high-density haze and clouds prevent light from penetrating, so the RGB band cannot capture the information behind this haze. Despite DehazeFormer [34] based on the transformer and the proposed method incorporating the latest approaches, the same problem occurred. Considering the recent advancements in generative models and the significantly improved performance of inpainting algorithms, it is conceivable that they could help ad-

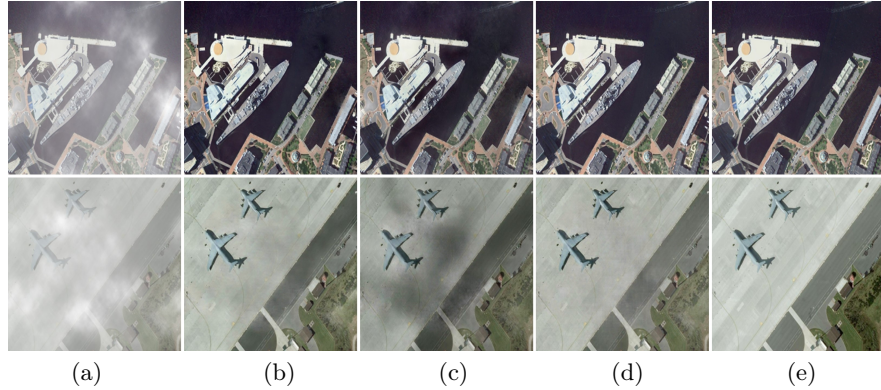


Fig. 9: Qualitative comparison using the RSID dataset [8]. (a) Input, (b) FFA-Net [31], (c) EMPF-Net [36], (d) Ours, (e) GT

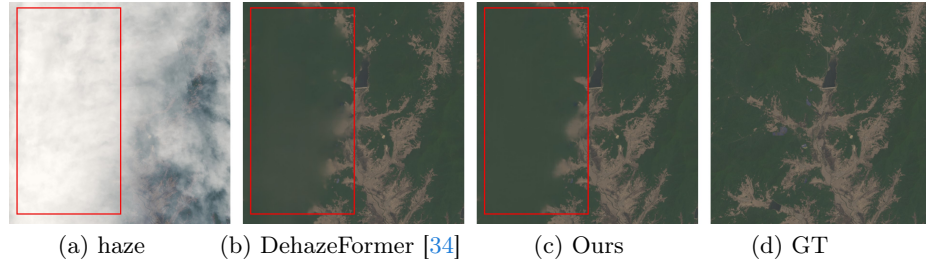


Fig. 10: Limitations of dense haze on satellite images.

dress these challenges. However, due to the characteristics of satellite images, delivering incorrect information due to content distortion can be highly risky, necessitating careful consideration. Other bands mounted on satellites, such as SAR, have the capability to penetrate clouds or haze and may offer assistance, but there are still challenges related to cost and the alignment of datasets. Also, the proposed method may not be easily applicable to real-world satellite haze images as it lacks the aspect of considering various satellites and various forms of haze. Therefore, we plan to improve these aspects in the future so that it can also be applied to real-world satellite images.

5 Conclusion

In this paper, we revisit the feasibility of employing masked image modeling for dehazing and utilize data augmentation through jigsaw transformation. To effectively apply masked image modeling to dehazing, we employed a small mask size and incorporated convolution. This approach demonstrated strengths in haze reconstruction and also exhibited improved performance in dehazing as a pre-trained model. Furthermore, the proposed approaches of masked image modeling and jigsaw transformation were shown to engage in complementary learning when utilized together, resulting in higher performance. Experimental results indicated that the proposed method achieved high performance in satellite image dehazing, both quantitatively and qualitatively, when compared to state-of-the-art algorithms.

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